

6CCSAHAI

Human AI Interaction

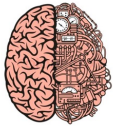
LECTURE 4 – LGT

Understanding Requirements in Human-AI systems

[What do we want to get done today ?]

1. **Deep-dive into interviewing techniques** [30 min interactive Q&A]
(... as a key method for user-needs elicitation)
2. **LGT-task: putting interviews into practice** [40 min]
 - design interview, use it on your colleagues, do an empathy map
3. **SGT-task : requirements engineering** [30 min + SGT]
 - develop a plan for your future system – what features? Why?

[mentimeter – both week 3 and week 4]



mental models
+ cognition



- **Effective** to use
- **Efficient** to use
- **Safe** to use
- Has good **utility**
- Easy to **learn**
- Easy to **remember**

2019

Guidelines for Human-AI Interaction

Salma Amari, Dan Wolf¹, Mikaela Vorvorean, Adam Feeney, Benjamin Nochi, Fouzi Collinette,
Jin Suh, Shant Jha, Paul N. Bennett, Kori Isken, Jaime Yeevan, David Kikio-Gil, and Eric Horvitz

Microsoft
Redmond, WA, USA
{samam, mivorv, adamf, bennochi, pncollin, jinsh,
shant, paulnb, kori, tovan, rishig, horvitz}
@microsoft.com

¹Paul G. Allen School of Computer
Science & Engineering
University of Washington
Seattle, WA, USA
wolfp@cs.washington.edu

2024

Design Principles for Generative AI Applications

Justin D. Weisz jweisz@us.ibm.com IBM Research AI Yorktown Heights, NY, USA	Jessica He jessicahe@ibm.com IBM Research AI Seattle, WA, USA	Michael Muller michael_muller@us.ibm.com IBM Research AI Cambridge, MA, USA
Gabriele Hofer ghofer@ibm.com IBM New York, NY, USA	Rachel Miles Rachel.Miles@ibm.com IBM San Jose, CA, USA	Werner Geyer Werner.Geyer@us.ibm.com IBM Research AI Cambridge, MA, USA

week 3

Human-centred AI as a Design Process

Part 1: Understanding user needs & Defining the right problem

Dr Georgia Panagiotidou



week 4

[theme of the day]

Engaging with humans (yuck!) ... aka 'empathise' and 'define'

*Walk in borrowed shoes.
Crystallize the hidden wound.
Design finds its soul.*

*Walk softly through others' despair,
Breathe the weight of the world that they bear.
Frame the question right,
Bring the need into light—
Innovation blooms from this care.*

Going from needs to requirements can be tricky

Myth of King Midas

Midas was a king of great fortune who ruled the country of Phrygia, in Asia Minor.

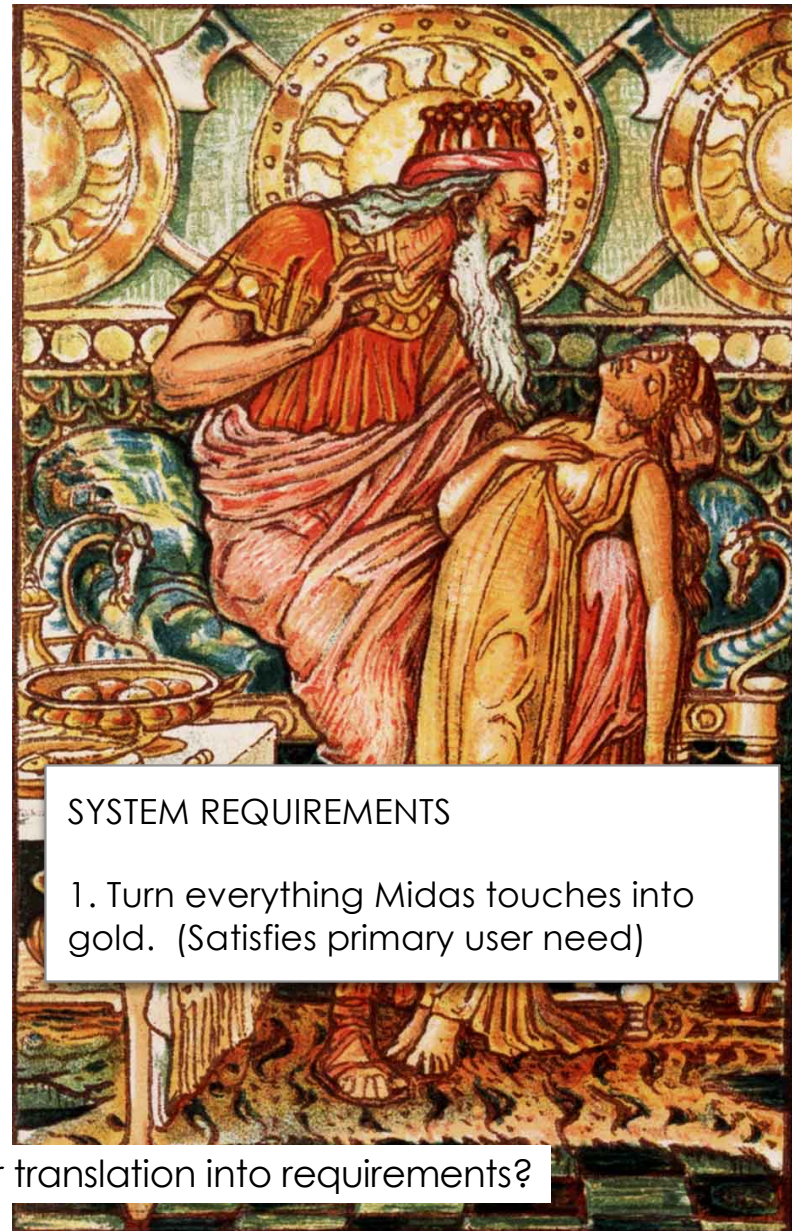
But his main pleasure was derived from gold. His avarice was such that he used to spend his days counting his golden coins! Occasionally he used to cover his body with gold objects, as if he wanted to bath in them. Money was his obsession.

(Source: www.greeka.com)

USER NEEDS

Who: King Midas

Needs: Gold. Loads and loads of it!
Happiness is derived from everything around him being made of pure gold.

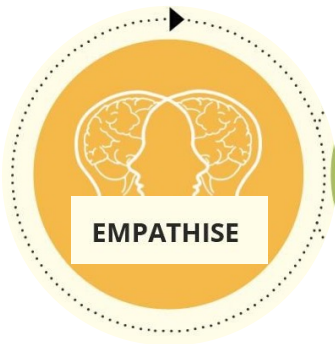


SYSTEM REQUIREMENTS

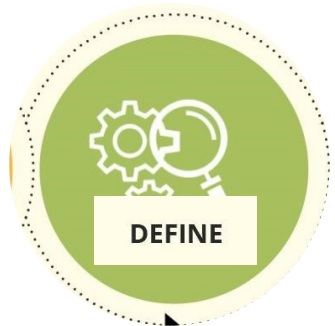
1. Turn everything Midas touches into gold. (Satisfies primary user need)

Failure in user needs elicitation or translation into requirements?

How do **needs** and **requirements** interact together?



1. To understand as much as possible about the **users**, their **activities**, and the **context** of that activity, so the system under development can support them in achieving their goals



2. To produce a **set of initial requirements** that form a sound basis to start designing

© MARK ANDERSON

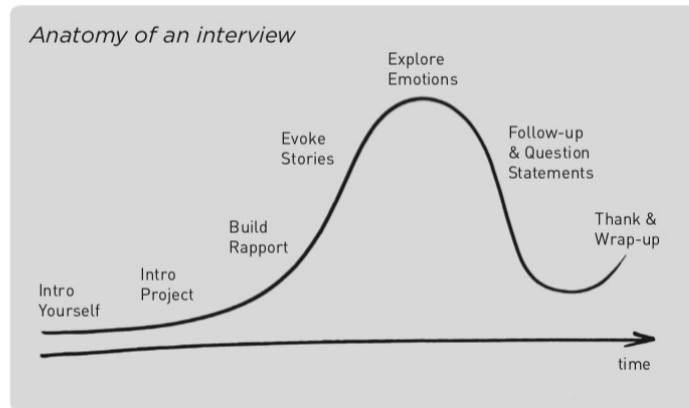
Can you list all
your needs please

WWW.ANDERSTOONS.COM

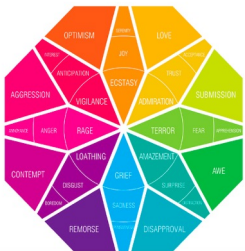


How to interview? Have a conversation.

Be human.
Seek **stories**.
Talk about **feelings**.



Adapted from Michael Barry



Be human: build rapport

Introduction: introduce yourself and your project in way that is comfortable to you. Try something like: “We’re working on a design project about the _____ experience. Could we talk to you for a few minutes?” Exchange names.

“How are you doing today?” (and actually listen).

Build trust: offer something of yourself. Be affirming.

“Tell me a little bit about (your experience today . . . , how you use . . . , what you think about . . .)”

Seek stories

Evoke **specific** stories to learn about what your interviewee does, and more importantly, thinks and feels. We are **talking about the past and present**, not the future.

“Could you tell me story about a time you _____?”

“What was your best/worst/craziest/most memorable experience with (area of focus).”

“What would I find surprising about . . .”

[write your own!]

Talk about feelings

“Walk me through how you . . . (made that decision, completed that task, got to a place, etc.). What were you thinking at that point?”

“Why do you say that? . . . “Tell me more.”

“How did you feel at that moment, when _____ happened?”

[write your own!]

Interviewing

DEFINITION

How do we know if we're actually solving users' needs? We talk to them! Interviewing is one of the most effective methods of Empathy. To effectively conduct interviews, we need to avoid Leading Questions, and we need to ask questions that move from Open-Ended to Specific to get the data we need. It is good to have a hunch of what your user(s) will say, but the goal of the interview is NOT to confirm our biases but to surprise us and enlighten us.

TIME: 30 minutes - 1 hour

MINDSETS

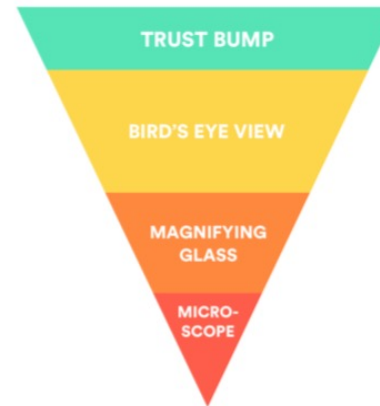
- Focus on Human Values

TIP: Remember to SCHEDULE interviews with the individuals chosen from Choosing Your Users. Reach out to those individuals (i.e. email), get a confirmation time, mark it in your calendar, and send a Google Calendar invite.

THE INTERVIEW LIFECYCLE

BROAD
(Start Here)

NARROW
(End Here)



Intent + Introductions
Tell me about your background in...

Tell me about the last time you...

Can you tell me more...
Can you show me how you...

5 Why's: Why do you...
Thank You's

TIPS FOR SUCCESS

- **Allow for Pregnant Pauses:** don't fill silences; give them time to answer
- **Stay Unbiased:** don't correct, judge, refute, or challenge
- **No Leading Questions:** don't ask "what problems do you have"; instead ask open-ended questions and allow them to tell you
- **No Hypotheticals:** don't ask "would you use this" because in politeness the interviewee often gives a shallow yes
- **Pair Interview:** interview with another Sprint member to avoid the distraction of multi-tasking

2. Framing questions: What questions are you going to ask?

You may find it helpful to draw up the following table to help you separate out what you want to find out and what you'll actually ask. It can sometimes be difficult for people to answer direct questions: try to find ways to ask questions that will allow your interview participant to speak naturally but at the same time give you the kinds of information or insights you need.

What I want to find out	Questions I will ask

As you develop your questions, these tips may be helpful.

- Avoid leading questions, or in other words, questions that steer your interview participant to give you the answers you are hoping to hear. Try to ask your questions as open-endedly as possible.
- Ask your interview participant to describe an experience ("Tell me about ..." or "What was it like for you to...?"). Try to avoid asking them directly to remember certain events because that can cause anxiety regarding the accuracy of their memories.
- Try asking interview participants to talk to you about a hypothetical situation or as if you or they were someone else – for example, "If I were a parent seeking advice about helping my child develop better reading habits, what would you say to me?" or "How would you explain the word "culture" to a child in 3rd grade?"
- Try to get your interview participant to tell a story about an experience.

Indirect questions: 'What do most people round here think of the ways that management treats its staff?', perhaps followed up by 'Is that the way you feel too?', in order to get at the individual's own view.

Follow-up questions: getting the interviewee to elaborate his/her answer, such as 'Could you say some more about that?'; 'What do you mean by that . . .?'

Probing questions: following up what has been said through direct questioning.

Specifying questions: 'What did you do then?'; 'How did X react to what you said?'

Interpreting questions: 'Do you mean that your leadership role has had to change from one of encouraging others to a more directive one?'; 'Is it fair to say that what you are suggesting is that you don't mind being friendly towards customers most of the time, but when they are unpleasant or demanding you find it more difficult?'

[our design brief]

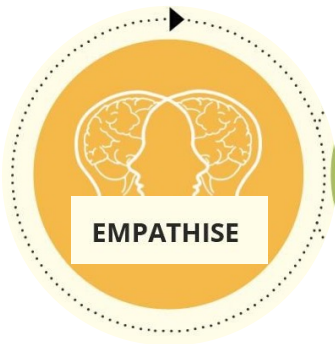
Design a system that will

- *guide students to likely set of challenges or difficulties with a learning material (e.g., crucial concepts that are likely to be on the exam);*
- *do so in an ideally personalised way—if possible; and*
- *then provide the students with an opportunity to test their knowledge (e.g., through a mock oral exam / job interview).*

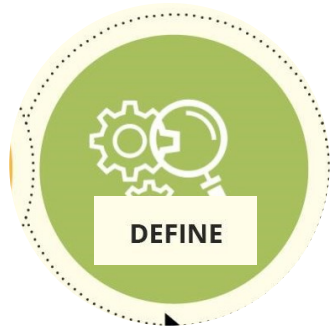
*Walk in borrowed shoes.
Crystallize the hidden wound.
Design finds its soul.*

*Walk softly through others' despair,
Breathe the weight of the world that they bear.
Frame the question right,
Bring the need into light—
Innovation blooms from this care.*

How do **needs** and **requirements** interact together?



1. To understand as much as possible about the **users**, their **activities**, and the **context** of that activity, so the system under development can support them in achieving their goals



2. To produce a **set of initial requirements** that form a sound basis to start designing

[LGT task – work in triads]

Your mission:

create an interview guide, try it on a human, learn & update

Step 1: Take up to 15 minutes to create an interview guide.

Your goals are to:

- Think about what questions to ask
- Test these on each other
- Update and write out your final interview guide

You will be testing the interview with a person from another team in Step 2!

[LGT task – work in triads]

Your mission:

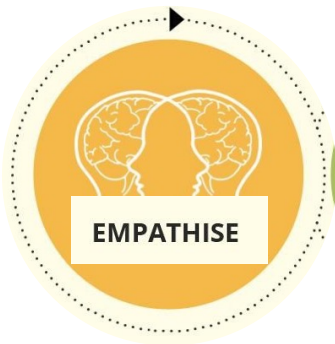
create an interview guide, try it on a human, learn & update

Step 2: Spend 10 minutes interviewing a colleague from another team
(+ send one member to be interviewed)

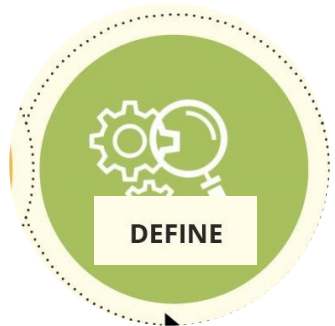
One of you should be doing the interviewing, the other keeps notes.

Step 3: Spend 15 minutes discussing your findings + welcome back comments from your returning team member.

How do **needs** and **requirements** interact together?

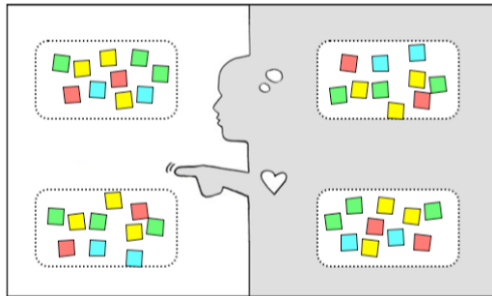


1. To understand as much as possible about the **users**, their **activities**, and the **context** of that activity, so the system under development can support them in achieving their goals



2. To produce a **set of initial requirements** that form a sound basis to start designing

METHOD EMPATHY MAP



WHY use an empathy map

Good design is grounded in a deep understanding of the person for whom you are designing. Designers have many techniques for developing this sort of empathy. An Empathy Map is one tool to help you synthesize your observations and draw out unexpected insights.

HOW to use an empathy map

UNPACK: Create a four quadrant layout on paper or a whiteboard. Populate the map by taking note of the following four traits of your user as you review your notes, audio, and video from your fieldwork:

SAY: What are some quotes and defining words your user said?
DO: What actions and behaviors did you notice?
THINK: What might your user be thinking? What does this tell you about his or her beliefs?
FEEL: What emotions might your subject be feeling?

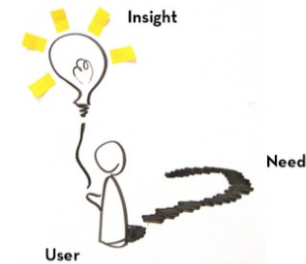
Note that thoughts/beliefs and feelings/emotions cannot be observed directly. They must be inferred by paying careful attention to various clues. Pay attention to body language, tone, and choice of words.

IDENTIFY NEEDS: "Needs" are human emotional or physical necessities. Needs help define your design challenge. Remember: Needs are *verbs* (activities and desires with which your user could use help), not *nouns* (solutions). Identify needs directly out of the user traits you noted, or from contradictions between two traits - such as a disconnect between what she says and what she does. Write down needs on the side of your Empathy Map.

IDENTIFY INSIGHTS: An "Insight" is a remarkable realization that you could leverage to better respond to a design challenge. Insights often grow from contradictions between two user attributes (either within a quadrant or from two different quadrants) or from asking yourself "Why?" when you notice strange behavior. Write down potential insights on the side of your Empathy Map. One way to identify the seeds of insights is to capture "tensions" and "contradictions" as you work.



METHOD POINT-OF-VIEW MADLIB



WHY use a POV madlib

A point-of-view (POV) is your reframing of a design challenge into an actionable problem statement that will launch you into generative ideation. A POV Madlib provides a scaffolding to develop your POV. A good POV will allow you to ideate in a directed manner, by creating How-Might-We (HMW) questions based on your POV (see "Facilitating Brainstorms"). Most of all, your POV captures your design vision - your responsibility and opportunity as a designer is to discover and articulate the meaningful challenge.

HOW to use a POV madlib

Use one of the following madlibs to communicate your new perspective on the challenge. Both versions focus your work by prompting you to choose a specific user, need, and useful insight.

[USER] needs to [USER'S NEED] because/but/Surprisingly [INSIGHT].

We met... [user]

We were amazed to discover... [what you noticed]

We wonder if this means... [insight]

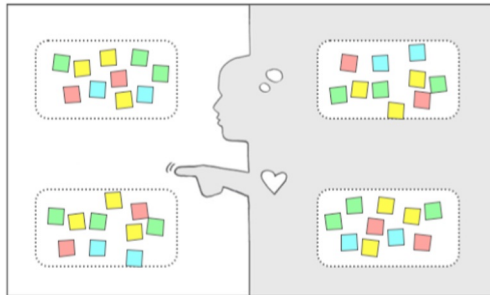
We aim to (help our user)... [need/goal]

Use a whiteboard or scratch paper to try out a number of options, playing with each variable and the combinations of them. Keep it specific (don't lose the nuance) and hold the tension in your POV.

For example, instead of "A teenage girl needs more nutritious food because vitamins are vital to good health" you might write "A teenage girl with a bleak outlook needs to feel more socially accepted when eating healthy food, because in her hood a social risk is more dangerous than a health risk." Note how the latter is an actionable, and potentially generative, problem statement, while the former is little more than a statement of fact, which spurs little inspiration or direction to develop solutions.



METHOD EMPATHY MAP



WHY use an empathy map

Good design is grounded in a deep understanding of the person for whom you are designing. Designers have many techniques for developing this sort of empathy. An Empathy Map is one tool to help you synthesize your observations and draw out unexpected insights.

HOW to use an empathy map

UNPACK: Create a four quadrant layout on paper following four traits of your user as you review your

SAY: What are some quotes and defining words your user said?
DO: What actions and behaviors did you notice?
THINK: What might your user be thinking? What does this tell you about his or her beliefs?
FEEL: What emotions might your subject be feeling?

Note that thoughts/beliefs and feelings/emotions cannot be observed directly. They must be inferred by paying careful attention to various clues. Pay attention to body language, tone, and choice of words.

IDENTIFY NEEDS: "Needs" are human emotional challenges. Remember: Needs are *verbs* (activities, nouns (solutions). Identify needs directly out of the two traits - such as a disconnect between what is said of your Empathy Map.

IDENTIFY INSIGHTS: An "Insight" is a remarkable design challenge. Insights often grow from contradictions (from two different quadrants) or from behavior. Write down potential insights on the side of insights is to capture "tensions" and "contradictions".

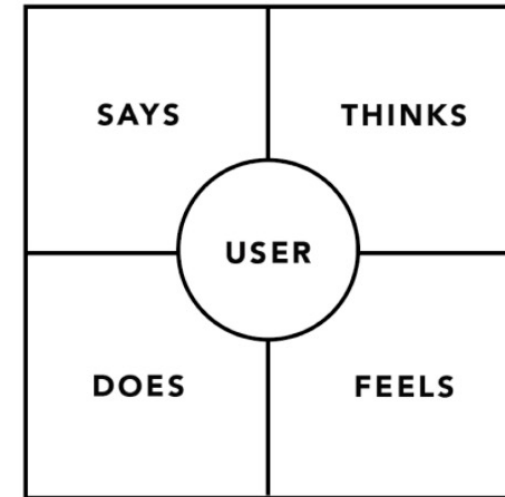
HOW to use an empathy map

UNPACK: Create a four quadrant layout on paper or a whiteboard. Populate the map by taking note of the following four traits of your user as you review your notes, audio, and video from your fieldwork:

SAY: What are some quotes and defining words your user said?
DO: What actions and behaviors did you notice?
THINK: What might your user be thinking? What does this tell you about his or her beliefs?
FEEL: What emotions might your subject be feeling?

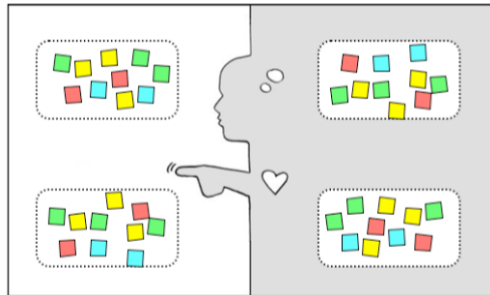
Note that thoughts/beliefs and feelings/emotions cannot be observed directly. They must be inferred by paying careful attention to various clues. Pay attention to body language, tone, and choice of words.

EMPATHY MAP



NNGROUP.COM **NN/g**

METHOD EMPATHY MAP



WHY use an empathy map

Good design is grounded in a deep understanding of the person for whom you are designing. Designers have many techniques for developing this sort of empathy. An Empathy Map is one tool to help you synthesize your observations and draw out unexpected insights.

HOW to use an empathy map

UNPACK: Create a four quadrant layout on paper or a whiteboard. Populate the map by taking note of the following four traits of your user as you review your notes, audio, and video from your fieldwork:

- SAY: What are some quotes and defining words your user said?
- DO: What actions and behaviors did you notice?
- THINK: What might your user be thinking? What does this tell you about his or her beliefs?
- FEEL: What emotions might your subject be feeling?

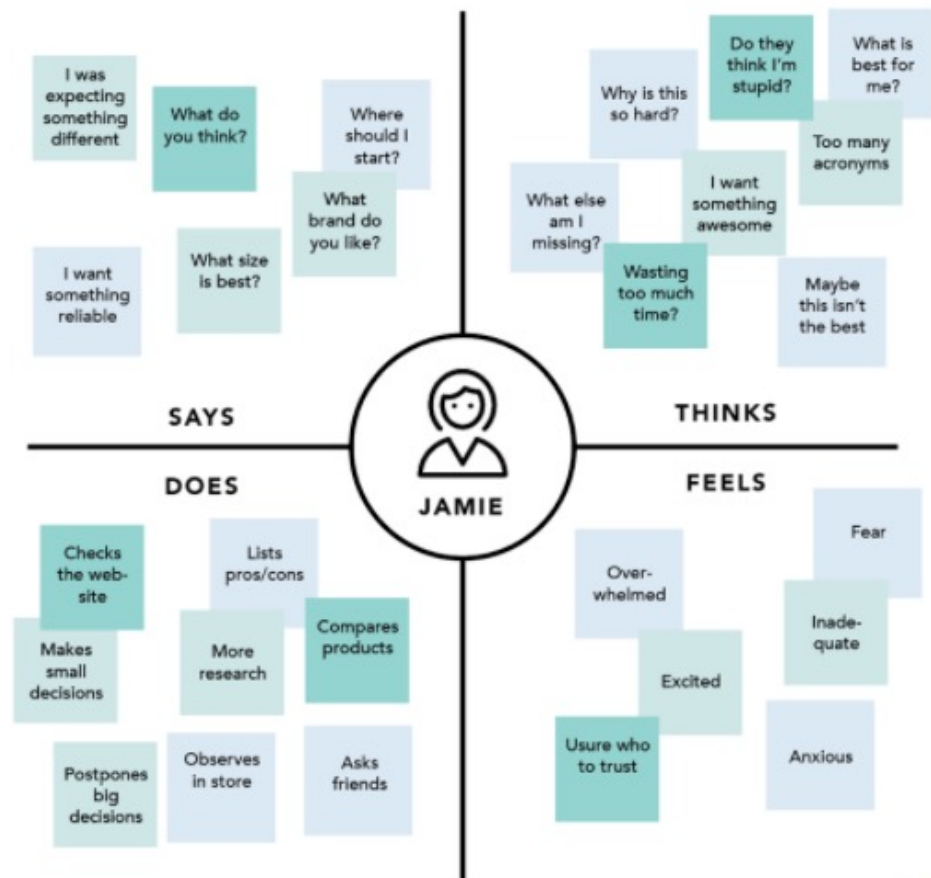
Note that thoughts/beliefs and feelings/emotions cannot be observed directly. They must be *inferred* by paying careful attention to various clues. Pay attention to body language, tone, and choice of words.

IDENTIFY NEEDS: "Needs" are human emotional or physical necessities. Needs help define your design challenge. Remember: Needs are *verbs* (activities and desires with which your user could use help), not *nouns* (solutions). Identify needs directly out of the user traits you noted, or from contradictions between two traits – such as a disconnect between what she says and what she does. Write down needs on the side of your Empathy Map.

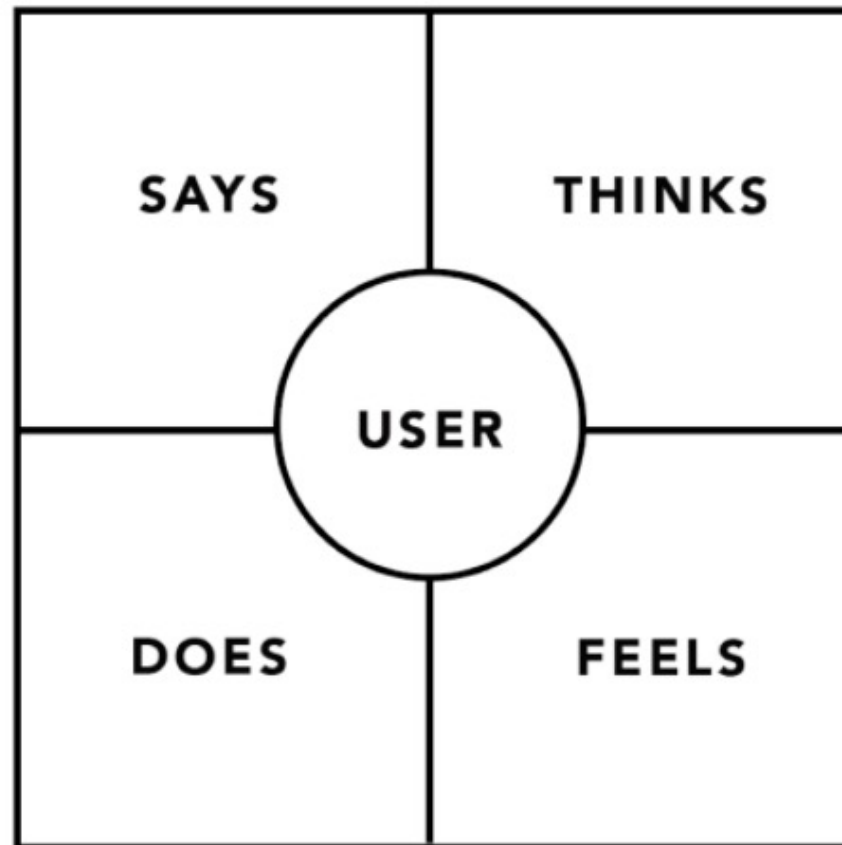
IDENTIFY INSIGHTS: An "Insight" is a remarkable realization that you could leverage to better respond to a design challenge. Insights often grow from contradictions between two user attributes (either within a quadrant or from two different quadrants) or from asking yourself "Why?" when you notice strange behavior. Write down potential insights on the side of your Empathy Map. One way to identify the seeds of insights is to capture "tensions" and "contradictions" as you work.

d.

EMPATHY MAP *Example (Buying a TV)*

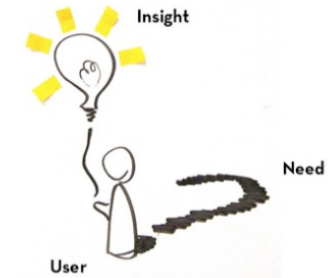


EMPATHY MAP



Point of view technique:

METHOD POINT-OF-VIEW MADLIB



WHY use a POV madlib

A point-of-view (POV) is your reframing of a design challenge into an actionable problem statement that will launch you into generative ideation. A POV Madlib provides a scaffolding to develop your POV. A good POV will allow you to ideate in a directed manner, by creating How-Might-We (HMW) questions based on your POV (see "Facilitating Brainstorms"). Most of all, your POV captures your design vision – your responsibility and opportunity as a designer is to discover and articulate the meaningful challenge.

HOW to use a POV madlib

Use one of the following madlibs to communicate your new perspective on the challenge. Both versions focus your work by prompting you to choose a specific user, need, and useful insight.

[USER] needs to [USER'S NEED] because/but/Surprisingly [INSIGHT].

We met . . . [user]

We were amazed to discover . . . [what you noticed]

We wonder if this means . . . [insight]

We aim to (help our user) . . . [need/goal]

Use a whiteboard or scratch paper to try out a number of options, playing with each variable and the combinations of them. Keep it specific (don't lose the nuance) and hold the tension in your POV.

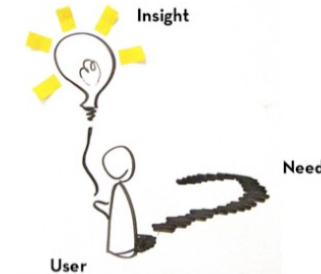
For example, instead of "A teenage girl needs more nutritious food because vitamins are vital to good health" you might write "A teenage girl with a bleak outlook needs to feel more socially accepted when eating healthy food, because in her hood a social risk is more dangerous than a health risk." Note how the latter is an actionable, and potentially generative, problem statement, while the former is little more than a statement of fact, which spurs little inspiration or direction to develop solutions.

Point of view technique:

WHY use a POV madlib

A point-of-view (POV) is your reframing of a design challenge into an actionable problem statement that will launch you into generative ideation. A POV Madlib provides a scaffolding to develop your POV. A good POV will allow you to ideate in a directed manner, by creating How-Might-We (HMW) questions based on your POV (see “Facilitating Brainstorms”). Most of all, your POV captures your design vision – your responsibility and opportunity as a designer is to discover and articulate the meaningful challenge.

METHOD POINT-OF-VIEW MADLIB



Use a whiteboard or scratch paper to try out a number of options, playing with each variable and the combinations of them. Keep it specific (don't lose the nuance) and hold the tension in your POV.

For example, instead of “A teenage girl needs more nutritious food because vitamins are vital to good health” you might write “A teenage girl with a bleak outlook needs to feel more socially accepted when eating healthy food, because in her hood a social risk is more dangerous than a health risk.” Note how the latter is an actionable, and potentially generative, problem statement, while the former is little more than a statement of fact, which spurs little inspiration or direction to develop solutions.

[SGT task – work in triads – ‘define’]

Your task will be to create an

- **empathy map**
- **Point-of-View matlib**
- **start brainstorming a set of features your system could have**

... this will be the basis for your prototyping / development work in the next week LGT + SGT!

... our design brief here ...

Role: Design team starting to explore the topic ^^

Output: One files to submit, which should include:

1. **Your interview guide**
2. **Empathy map (e.g., a photo) + Point-of-view**
3. **Any initial thoughts on your future system design**

[SGT task – work in pairs / triads]

Suggested time progression:

- spend **10-20 minutes** on making the empathy map ,
- then **15-20 minutes** on ideating a few Point-of-View statements, and then select one
- **rest of time** to finalise & write up your SGT submission

1. Classic d.school POV Mad Lib

[User] needs a way to [user's need] because [surprising insight].

Example:

A first-time college student needs a way to feel connected on campus because orientation week feels overwhelming and impersonal.

2. Emotion-Focused POV

[User] feels [emotion] when [situation], and needs a way to [user's need] so they can [desired emotional outcome].

Example:

A remote worker feels isolated when working alone all week, and needs a way to build small social connections so they can feel part of a team.

3. Constraint-Focused POV

[User] needs a way to [user's need], but is limited by [constraint/barrier], because [insight about why this matters].

Example:

A busy parent needs a way to exercise regularly, but is limited by childcare responsibilities, because they can't find safe, flexible fitness options at home.

4. Behavior-Focused POV

[User] is trying to [behavior or goal], but struggles because [barrier or misunderstanding], so they need a way to [help them succeed].

Example:

A commuter is trying to bike to work, but struggles because there's no safe storage for their bike, so they need a way to store it securely near the office.

5. Context + Insight POV

In the context of [situation or environment], [user] needs [need] because [human insight about motivation or behavior].

Example:

In the context of hospital waiting rooms, family members of patients need reassurance and updates because uncertainty amplifies their stress and helplessness.

